Class 10: Structural Bioinformatics 1

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PDB Statistics

The Protein Data Bank (PDB) is the main repository of biomolecular strucutres. Let's see what it contains:

```
stats <- read.csv("Data Export Summary.csv")
stats
```

```
##            Molecular.Type   X.ray     EM    NMR Integrative Multiple.methods
## 1          Protein (only) 178,795 21,825 12,773         343              226
## 2 Protein/Oligosaccharide  10,363  3,564     34           8               11
## 3              Protein/NA   9,106  6,335    287          24                7
## 4     Nucleic acid (only)   3,132    221  1,566           3               15
## 5                   Other     175     25     33           4                0
## 6  Oligosaccharide (only)      11      0      6           0                1
##   Neutron Other   Total
## 1      84    32 214,078
## 2       1     0  13,981
## 3       0     0  15,759
## 4       3     1   4,941
## 5       0     0     237
## 6       0     4      22
```

```
stats$X.ray
```

```
## [1] "178,795" "10,363"  "9,106"   "3,132"   "175"     "11"
```

```
sum(stats$Neutron)
```

```
## [1] 88
```

The comma in these numbers leads to the numbers here bring read as character.

```
c("100", "10", "barry")
```

```
## [1] "100"   "10"    "barry"
```

```
#install.packages("readr")
library(readr)
stats <- read_csv("Data Export Summary.csv")
```

```
## Rows: 6 Columns: 9
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr (1): Molecular Type
## dbl (4): Integrative, Multiple methods, Neutron, Other
## num (4): X-ray, EM, NMR, Total
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
stats
```

```
## # A tibble: 6 × 9
##   `Molecular Type`    `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
##   <chr>                 <dbl> <dbl> <dbl>       <dbl>              <dbl>   <dbl>
## 1 Protein (only)       178795 21825 12773         343                226      84
## 2 Protein/Oligosacch…   10363  3564    34           8                 11       1
## 3 Protein/NA             9106  6335   287          24                  7       0
## 4 Nucleic acid (only)    3132   221  1566           3                 15       3
## 5 Other                   175    25    33           4                  0       0
## 6 Oligosaccharide (o…      11     0     6           0                  1       0
## # ℹ 2 more variables: Other <dbl>, Total <dbl>
```

```
n.xray <- sum(stats$`X-ray`)
#n.em <-
n.total <- sum(stats$Total)

n.xray/n.total
```

```
## [1] 0.8095077
```

> Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
n.xray <- sum(stats$`X-ray`)
n.em <- sum(stats$`EM`)
n.total <- sum(stats$Total)

(n.xray + n.em)/n.total
```

```
## [1] 0.937892
```

> Q2: What proportion of structures in the PDB are protein?

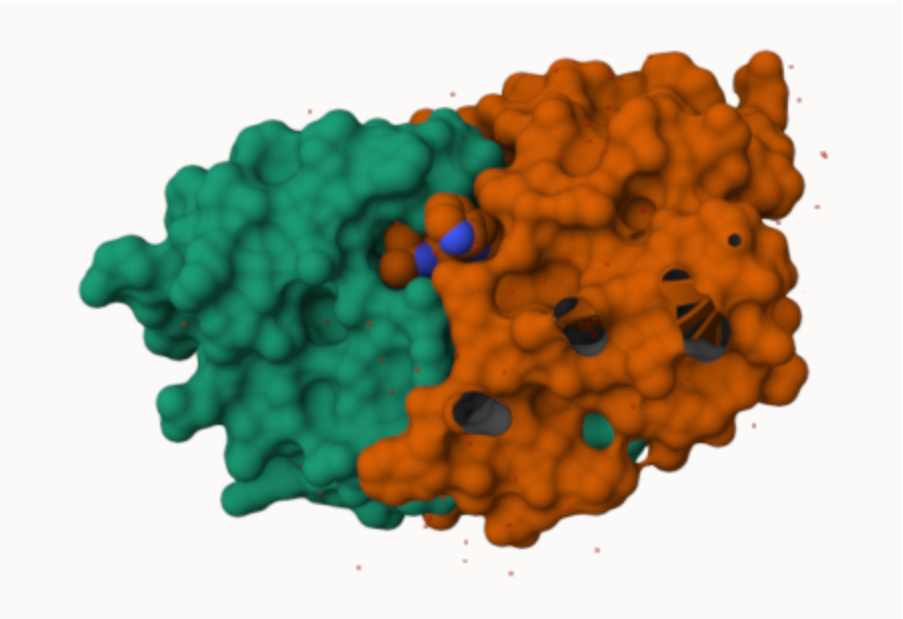
```
n.protein <- sum(stats[1,9])
n.protein/n.total
```

```
## [1] 0.8596889
```

> Q3: SKIP… Looking up HIV structures including 1HSG

Visualizing the HIV-1 protease structure

We can use the Molstar viewer online: https://molstar.org/viewer/ (https://molstar.org/viewer/)



My first image of HIV-Pr with surface display showing ligand binding
A new clean image showing the catalytic ASP25 amino acids in both chaings of the HIV-Pr dimer along with the inhibitor and all important active site water.

Bio3D package for structural bioinformatics

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

```
##   Note: Accessing on-line PDB file
```

```
pdb
```

```
##
##  Call:  read.pdb(file = "1hsg")
##
##    Total Models#: 1
##      Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)
##
##      Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
##      Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)
##
##      Non-protein/nucleic Atoms#: 172  (residues: 128)
##      Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
##
##    Protein sequence:
##       PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
##       QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
##       ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
##       VNIIGRNLLTQIGCTLNF
##
## + attr: atom, xyz, seqres, helix, sheet,
##         calpha, remark, call
```

```
head(pdb$atom)
```

```
##   type eleno elety  alt resid chain resno insert      x      y     z o     b
## 1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
## 2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
## 3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
## 4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
## 5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
## 6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
##   segid elesy charge
## 1  <NA>     N   <NA>
## 2  <NA>     C   <NA>
## 3  <NA>     C   <NA>
## 4  <NA>     O   <NA>
## 5  <NA>     C   <NA>
## 6  <NA>     C   <NA>
```

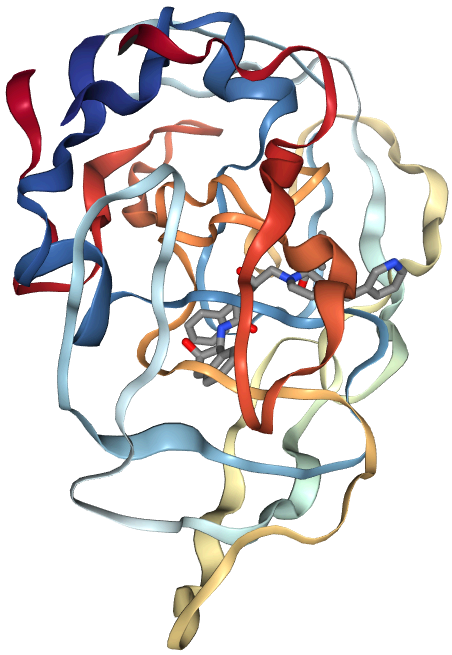
```
#install.packages("pak")
pak::pak("bioboot/bio3dview")
```

```
## ! Using bundled GitHub PAT. Please add your own PAT using `gitcreds::gitcreds_set()`.
```

```
## ℹ Loading metadata database✔ Loading metadata database ... done
##
## ℹ No downloads are needed
## ✔ 1 pkg + 40 deps: kept 37 [7.5s]
```

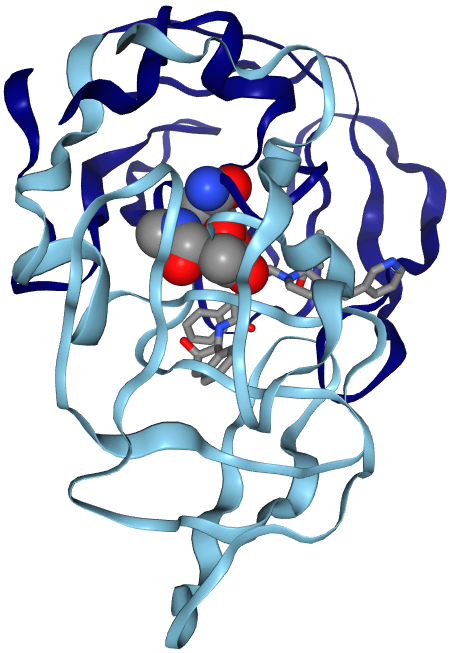
```
library(bio3dview)

view.pdb(pdb)
```



```
# Select the important ASP 25 residue
sele <- atom.select(pdb, resno=25)

# and highlight them in spacefill representation
view.pdb(pdb, cols=c("navy","skyblue"),
         highlight = sele,
         highlight.style = "spacefill")
```



Predicting functional motions of a single structure

Read an ADK structure from the PDB database:

```
adk <- read.pdb("6s36")
```

```
##   Note: Accessing on-line PDB file
##    PDB has ALT records, taking A only, rm.alt=TRUE
```

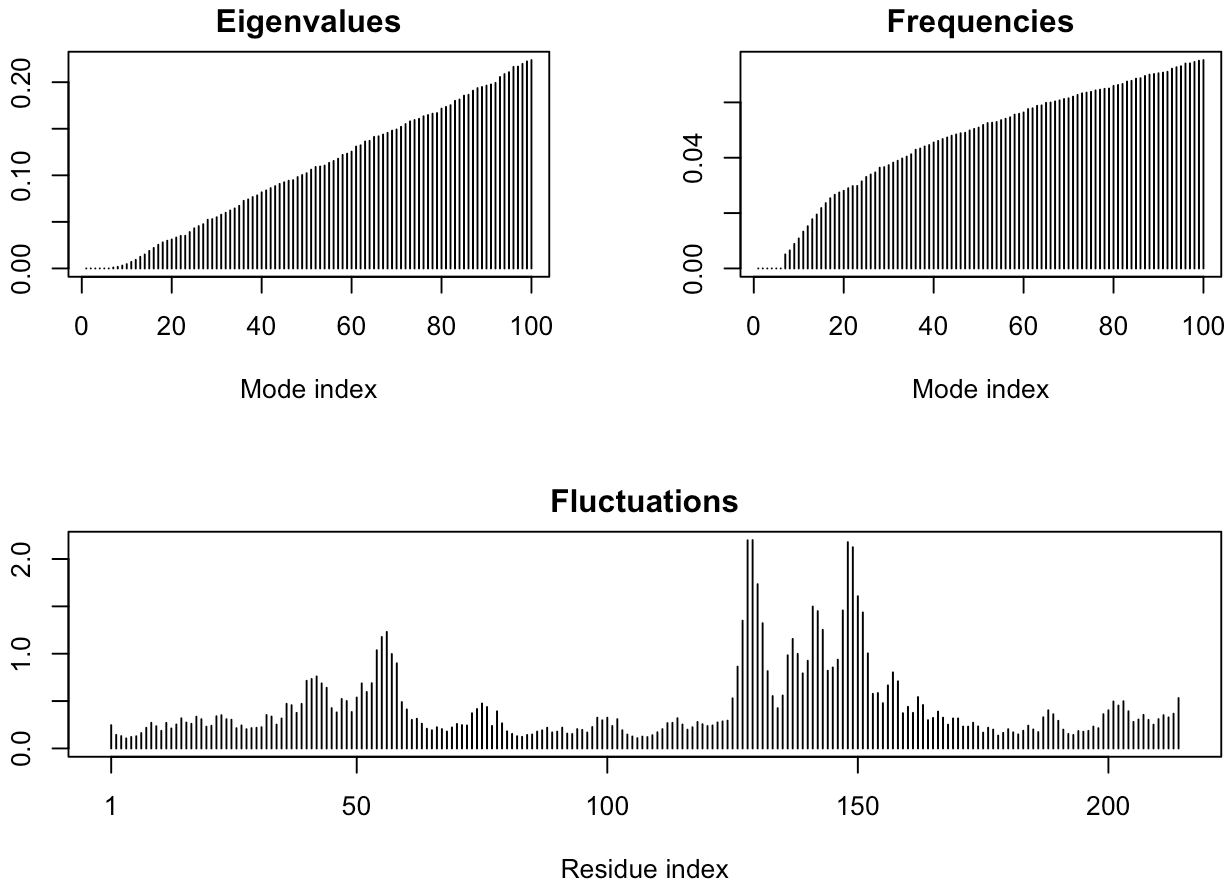
```
adk
```

```
##
##  Call:  read.pdb(file = "6s36")
##
##    Total Models#: 1
##      Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)
##
##      Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
##      Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)
##
##      Non-protein/nucleic Atoms#: 244  (residues: 244)
##      Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
##
##    Protein sequence:
##       MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
##       DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
##       VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
##       YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
##
## + attr: atom, xyz, seqres, helix, sheet,
##         calpha, remark, call
```

```
m <- nma(adk)
```

```
##  Building Hessian...     Done in 0.019 seconds.
##  Diagonalizing Hessian...    Done in 0.093 seconds.
```

```
plot(m)
```



Write out our results as a wee ttrajectory/movie of predicted motions:

```
mktrj(m, file="adk_m7.pdb")
```